AI PRODUCT MANAGEMENT 2 · DETAILED COURSE

AI Product Evaluation, Testing and Observability

How to know whether an AI product is good, safe and staying good over time.

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CHAPTER 17

AI Product Evaluation, Testing and Observability

How to know whether an AI product is good, safe and staying good over time.

Traditional Software Testing vs AI Testing

Traditional tests expect exact outputs; AI outputs vary, so you score quality statistically on many cases.

Traditional tests expect exact outputs for given inputs. AI outputs vary, so quality is scored statistically across many cases, with a pass threshold set in advance.

AI Product Evaluation Fundamentals

Define what good means for users, build test sets, pick metrics and set a launch bar.

Define what good means to users, build representative test sets, choose metrics and set a launch bar. Without a definition of good, every release becomes an argument.

Offline Evaluation

Scoring a model or prompt on a fixed dataset before release.

Score a model or prompt on a fixed dataset before release. It is fast and repeatable, but it may not reflect real users.

Online Evaluation

Measuring real behaviour after release through experiments and user signals.

Measure real behaviour after release through experiments and user signals such as edits, retries and ratings. It shows the impact that matters, but carries user risk.

Human Evaluation

Reviewers rate outputs against a rubric; the gold standard but slow and costly.

Reviewers rate outputs against a rubric. It is the most trusted method but slow and expensive, so use it to calibrate automated scoring.

Automated Evaluation

Programmatic checks and model-based scoring that run quickly at scale.

Programmatic checks and model-based scoring run quickly at scale. Check the automated scores against human ratings regularly.

LLM-as-a-Judge

Using a model to grade outputs against a rubric; validate its agreement with human labels first.

A model grades outputs against a rubric. Validate its agreement with human labels first, and watch for bias toward longer or similar-sounding answers.

Golden Datasets

Curated, trusted examples with expected answers that anchor evaluation.

Curated, trusted examples with expected answers anchor evaluation. Keep them representative and update them when the product or users change.

Evaluation Datasets

Representative collections of inputs, including hard and adversarial cases, versioned and refreshed.

Representative collections of inputs, including hard and adversarial cases, are versioned and refreshed. A dataset that never changes stops finding new failures.

Benchmarking

Comparing models or versions on shared tasks; treat public benchmarks as a hint and test on your own data.

Benchmarking compares models or versions on shared tasks. Public benchmarks are a hint; test on your own data before deciding.

Accuracy and Relevance

Whether the answer is correct and addresses the question.

Accuracy asks whether the answer is correct. Relevance asks whether it addresses the question. An answer can be accurate but irrelevant.

Faithfulness

Whether claims are supported by the provided sources.

Faithfulness checks that claims are supported by the provided sources. It is the key quality check for grounded answers.

Hallucination Measurement

The rate of unsupported or false statements, tracked on a fixed test set.

Track the rate of unsupported or false statements on a fixed test set. Measure it on every model or prompt change.

Retrieval Evaluation

Checking whether the right passages are found, using precision, recall and ranking metrics.

Check whether the right passages are found, using precision, recall and ranking measures. Retrieval errors look like generation errors unless measured separately.

Agent Success Rate

The share of tasks an agent completes correctly end to end.

The share of tasks an agent completes correctly end to end. Break failures down by step to see where the agent struggles.

Task Completion Rate

The share of user tasks finished, with or without assistance.

The share of user tasks finished, with or without help. A low rate with high satisfaction can mean the task is too hard to finish.

Safety Evaluation

Testing for harmful, unsafe or policy-violating outputs.

Test for harmful, unsafe or policy-violating outputs using adversarial and everyday inputs. Safety tests should run before every release.

Toxicity and Bias Testing

Measuring offensive output and unequal treatment across groups.

Measure offensive output and unequal treatment across groups, using paired test inputs that differ only in the group mentioned.

Robustness Testing

Checking behaviour under typos, paraphrases, noise and adversarial inputs.

Check behaviour under typos, paraphrases, noisy inputs and adversarial prompts. A product that works only on perfectly written questions will fail in use.

Regression Testing

Re-running the evaluation suite after every change to catch quality drops.

Re-run the evaluation suite after every change to catch quality drops. Regressions are often silent until a user reports them.

AI Red Teaming

Deliberately attacking the system to find failures before attackers or users do.

Deliberately attack the system to find failures before attackers or users do. Include prompt injection, data extraction and misuse scenarios.

A/B Testing for AI

Comparing model or prompt variants on users, with quality, cost and safety guardrails.

Compare model or prompt variants on real users with quality, cost and safety guardrails. Keep the metrics defined before the test starts.

Model Latency

Response time measured at median and tail percentiles against a target.

Measure response time at median and tail percentiles against a target. Tail latency (the slowest 5 to 1 per cent) often drives user complaints.

Throughput

Requests handled per second under load.

Throughput is requests handled per second under load. Load test before launch so you know where the system slows down.

Cost per Request

Total model and infrastructure cost divided by requests; track per successful task as well.

Total model and infrastructure cost divided by requests. Also track cost per successful task, which is the number that shows real efficiency.

Token Consumption

Input and output tokens used, a driver of cost and latency.

Input and output tokens drive cost and latency. Monitor token use per feature to find prompts that grew without anyone noticing.

AI Observability

Logs, traces and metrics that explain what the system did and why.

Logs, traces and metrics explain what the system did and why. Observability turns a vague complaint into a specific failing step.

Production Monitoring

Dashboards and alerts on quality, errors, latency, cost and safety in live traffic.

Dashboards and alerts track quality, errors, latency, cost and safety in live traffic. Alerts should point to an owner and a runbook.

Model Drift

Performance changes as data, users or models change over time.

Performance changes as data, user behaviour or the model itself changes. Compare recent inputs with the data the system was tested on to detect drift early.

Continuous Evaluation

Ongoing scoring of live samples and test suites so quality is tracked, not assumed.

Score live samples and run the test suite continuously, so quality is tracked, not assumed. Quality can fall slowly, and continuous scoring is the only reliable way to notice.

Summary of terms

Term	Meaning
Traditional Software Testing vs AI Testing	Traditional tests expect exact outputs; AI outputs vary, so you score quality statistically on many cases.
AI Product Evaluation Fundamentals	Define what good means for users, build test sets, pick metrics and set a launch bar.
Offline Evaluation	Scoring a model or prompt on a fixed dataset before release.
Online Evaluation	Measuring real behaviour after release through experiments and user signals.
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LLM-as-a-Judge	Using a model to grade outputs against a rubric; validate its agreement with human labels first.
Golden Datasets	Curated, trusted examples with expected answers that anchor evaluation.
Evaluation Datasets	Representative collections of inputs, including hard and adversarial cases, versioned and refreshed.
Benchmarking	Comparing models or versions on shared tasks; treat public benchmarks as a hint and test on your own data.
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Regression Testing	Re-running the evaluation suite after every change to catch quality drops.
AI Red Teaming	Deliberately attacking the system to find failures before attackers or users do.
A/B Testing for AI	Comparing model or prompt variants on users, with quality, cost and safety guardrails.
Model Latency	Response time measured at median and tail percentiles against a target.
Throughput	Requests handled per second under load.
Cost per Request	Total model and infrastructure cost divided by requests; track per successful task as well.
Token Consumption	Input and output tokens used, a driver of cost and latency.
AI Observability	Logs, traces and metrics that explain what the system did and why.
Production Monitoring	Dashboards and alerts on quality, errors, latency, cost and safety in live traffic.
Model Drift	Performance changes as data, users or models change over time.

Continuous Evaluation	Ongoing scoring of live samples and test suites so quality is tracked, not assumed.

Practice exercise: Create a golden set of fifty cases for one AI feature, define a rubric, and score two prompt versions.

Self-check

1. Explain Traditional Software Testing vs AI Testing in two sentences to a non-technical stakeholder, then say what you would measure for it.
2. Explain Task Completion Rate in two sentences to a non-technical stakeholder, then say what you would measure for it.
3. Explain Continuous Evaluation in two sentences to a non-technical stakeholder, then say what you would measure for it.

Related SkillNavora material

- Agent Orchestration and Monitoring: Orchestrating and observing agents.
- AI Evals for Product Managers: Evaluation methods and datasets.
- AI Skills Academy: Every AI skills course in one place.
- AI Tutor: Ask questions on any topic.
- Study Library: Every lesson, class and PDF by programme.

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